VOLUME Appendix

# Appendix: Reproduction Protocols

Environment, seeds, data dictionary, control register, scorecard

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# Environment and determinism

All campaigns ran in one environment: Python 3.12, numpy 2.1, scipy 1.14 (eigvalsh, driver evr), mpmath 1.3 (dps 25–40 per criterion), matplotlib 3.9 (600 dpi figures). The master seed of the series is 96; the RNG streams are fixed: vortices — default_rng(96 + 3500), disorder — default_rng(96 + 3501), GUE nulls — default_rng(9000 + k), Poisson nulls — default_rng(7000 + k). Runtimes: C1 ~2 s, C2 ~30 min (two arithmetics), C3 ~3 min, C4 ~12 s, C5 ~5 s, C6 ~8 min (the $10^6$ sieve), C7 ~20 min (L=96 dynamics), C8 ~1 min.

The bit-for-bit base control: the protocol reference gate reproduces the mean level spacing of the C3 CSV with deviation 1.45e-16 — the kernel port (engine/hamiltonian.py) is equivalent to the original protocol up to summation order. All verdicts are committed in data/; the campaigns re-read them and recompute cheap invariants in seconds.

# Data dictionary

The ζ zero file (2 001 052 heights, Odlyzko format) is not shipped for licensing reasons; its path is set by ZFILE in campaigns/ and defaults to verification/data/zeros6.txt of the neighbouring repository. Every verdict of the series is committed, so no test requires the zero file; fresh campaigns with recomputation are optional.

*Table 1. Verdict files*

| File | Content | Campaign |
|---|---|---|
| c1_identity_verdict.json | K1–K7: ladder, reality, zeros, Gram | C1 |
| c2_argument_verdict.json | anchors 0/3/10/29/79/649, S(T), gaps, bridge | C2 |
| c3_gates_verdict.json + c3_gates_table.csv | 52 gates, nulls, PC1/PC2 | C3 |
| c4_trace_verdict.json | M0/M1/M2, PC0, trace-metric gap | C4 |
| c5_gates_ext_verdict.json + c5_gates_ext_table.csv | 35+2 extended gates | C5 |
| c6_primes_verdict.json | co-location, kernel, ψ-ladder, codings | C6 |
| c7_dynamics_verdict.json | trajectories ⟨x⟩(t), transport P(x≥90) | C7 |
| C8 ladder + spinor64 | 11-channel ladder, 64→4 reduction | C8 |

# Register of positive and negative controls

The discipline of controls: a method is allowed a negative verdict only after demonstrating a positive result on an object where identity is known to hold. Every negative result of the series (H2, H5, M3) passed this procedure; every positive one (T1–T4, T6, T8) passed an independent recount in IVP01–IVP10.

*Table 2. Controls of the series*

| Control | Must show | Value | Where |
|---|---|---|---|
| PC1 spinor64 twins | r = 1 on an isospectral orbit | r = 1.000000000, D_nn = 2.7e-14 | C3 |
| PC2 graph ↔ file | r = 1 between Z_id zeros and the file | r = 1.000000000000 | C3 |
| PC0 zeros window (trace) | the method sees the Λ-comb | r_EF = +0.9998, residual 2.1% | C4 |
| Decoys (random phases) | the null zone of correlations | 0.026±0.140 / +0.050±0.153 | C3, C6 |
| GUE null | universality without identity | ⟨r⟩-plateau; r_EF 0.43±0.71 | C3–C5 |
| Surrogate null (blocks) | the width of the trace metric | 0.00±0.39 | C4 |
| Poisson null | clustered spectra | 0.08±0.80 | C4 |

# The Hilbert–Pólya scorecard

*Table 3. The five questions of the programme*

| Item | Question | Verdict | Basis |
|---|---|---|---|
| H1 | self-adjointness of the operator | YES | Hermitian by construction; real spectrum |
| H2 | spectral identity level↔zero | NO | null level — D_nn 0.0079 vs surrogate 0.0079±0.0005 (C3); ≈0.5 order by relative residual (C4); \|r_H7\| ≤ 0.043 (C5) |
| H3 | the GUE class | YES | ⟨r⟩ = 0.60 at Nv ≥ 8; all of C5 in class |
| H4 | Dirac dynamics | YES | coherent packet, energy 1e-15 (C7) |
| H5 | primes in the operator | NO | ψ-ladder 0.972 for zeros, null zone for gates (C6) |

The cycle formula after eight campaigns: the resonator is built and tuned (H1+H3+H4); the zeros are encoded, not generated (H2+H5); the prime weight is verified only for the zero operator (T6). The roadmap — in results/ROADMAP.md: stitching the statistical and coding branches, nonlocal trace constructions, a number-theoretic explanation of T9.